JULKAR NIENE

System Engineer | *Linux · Nginx · MariaDB · Bash · Python*

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PROFESSIONAL SUMMARY

CSE graduate (Data Science major) moving into systems engineering. System Engineer intern with hands-on Linux experience on a lab server: hosting Angular, React, PHP and Laravel apps on Nginx and PHP-FPM, running multiple PHP versions side by side, configuring reverse proxies, and automating MariaDB backups with Bash and systemd timers. Comfortable with users, permissions, systemd, Bash scripting and cron. A full-stack background speeds up my debugging and root-cause tracing; I have deployed several applications (Vercel, Render, Railway, Neon) and write clear READMEs and API specs.

TECHNICAL SKILLS

Systems & Servers	Linux (Ubuntu Server), Nginx (web hosting, reverse proxy), PHP-FPM, systemd (services, timers, journalctl), users, groups & permissions, sudo, process and resource monitoring, archiving, text processing (grep, sed, awk)
Networking	Cisco Networking Basics, TCP/IP fundamentals, ip, ss, ping, curl, wget, SSH key authentication, scp
Scripting & Automation	Bash, Python, cron, systemd timers, n8n
Databases	MariaDB (mariadb-dump, restore, binary logging and point-in-time recovery concepts), SQL for administration (joins, aggregation, EXPLAIN/ANALYZE, query inspection), PostgreSQL
Deployment & Hosting	Nginx, PHP-FPM, Railway, Render, Vercel, Neon
Development	JavaScript, React.js, Next.js 14, FastAPI, Node.js/Express, REST API, JWT auth, Socket.io, Prisma ORM
Tooling	Git/GitHub (branching, merging, remotes over SSH), Postman, Jira, Swagger/OpenAPI

WORK EXPERIENCE

System Engineer Intern — Schertech (SCT Bangla) *Sep 2026 – Present*

- Deployed Angular, React (Vite), PHP and Laravel applications on Ubuntu Server using Nginx and PHP-FPM; installed dependencies, built production bundles, and resolved build errors.
- Configured multiple PHP versions to run concurrently as separate PHP-FPM services, routing each site to its required version through Nginx so legacy applications could run without disrupting newer projects.
- Configured Nginx as a reverse proxy to forward a path to a local backend application, handling proxy headers and path-prefix behaviour.
- Completed a MariaDB backup and restore exercise: hosted a PHP application from GitHub through Nginx, dumped its database with mariadb-dump, restored it into a separate database, and repointed the application configuration to the restored copy.

SYSTEMS PROJECTS

Linux Server Health Monitor

Bash · systemd timers · Nginx · Ubuntu Server

- Bash monitor that checks CPU, RAM, disk, uptime, SSH and Nginx, run on a schedule by systemd timers instead of cron.
- Sends an alert only when a status changes, and a RECOVERED message when the problem clears, to avoid repeated alerts.
- Publishes a status page through Nginx on port 8080.

MariaDB Manager

Bash · MariaDB · mariadb-dump · gzip · systemd timers

- Runs through a systemd service and timer (Persistent=true, so missed runs catch up after downtime); output is logged to a file and visible in journalctl.
- Bash tool to back up, restore, list and drop MariaDB databases using mariadb-dump (`--single-transaction`), with compressed dumps, gzip integrity checks after each backup, and automatic 7-day retention cleanup.
- Reads credentials from a protected `.my.cnf` instead of hardcoding them; adds confirmation prompts, safeguards against dropping system databases, timestamped logging, and a README with troubleshooting and known limits.

DEVELOPMENT PROJECTS

Collaborative Team Hub — Next.js 14 · Express · PostgreSQL/Prisma · Socket.io · Railway

- Built and deployed a full-stack monorepo to Railway under an assessment deadline, with JWT auth, role-based access

control, real-time messaging and Swagger/OpenAPI docs.

- Traced schema migration issues and race conditions in real-time event handling to root cause and documented the fixes.

Aria – Full Stack AI Assistant — Next.js · TypeScript · FastAPI · PostgreSQL · Redis

- Deployed across Vercel (frontend), Render (FastAPI backend), Neon (PostgreSQL) and Upstash (Redis for background jobs); implemented JWT and Google OAuth with email verification and a persistent WebSocket layer.

AI Job Application Tracking System — React · FastAPI · Supabase/PostgreSQL · Groq API

- Built a pipeline that parses job descriptions with an LLM API into a normalised PostgreSQL schema, with CV parsing (PDF/DOCX) that handles malformed PDFs and inconsistent formats.

EDUCATION

B.Sc in Computer Science & Engineering — United International University (UIU) *2022 – 2026 | Graduated*
Major: Data Science | CGPA: 3.4 / 4.00

ACHIEVEMENTS & CERTIFICATIONS

- Cisco Networking Academy — CCNA: Introduction to Networks
- Cisco Networking Academy — Networking Basics
- Cisco Networking Academy — Introduction to Cybersecurity
- 3rd Runner Up — UIU CSE Project Show (Database Management System)

LEADERSHIP & EXPERIENCE

Co-Head of HR — UIU App Forum (University Tech Club) *Sep 2024 – Oct 2025*
Led recruitment for a 100+ member tech club: defined roles, screened candidates, ran interviews and onboarded members.

Grader — United International University *Jun 2023 – Oct 2023*
Evaluated programming assignments and exams; gave structured written feedback and kept grading consistent with faculty.